WASTE MANAGEMENT INNOVATIONS

Waste Classification and Adversarial Robustness

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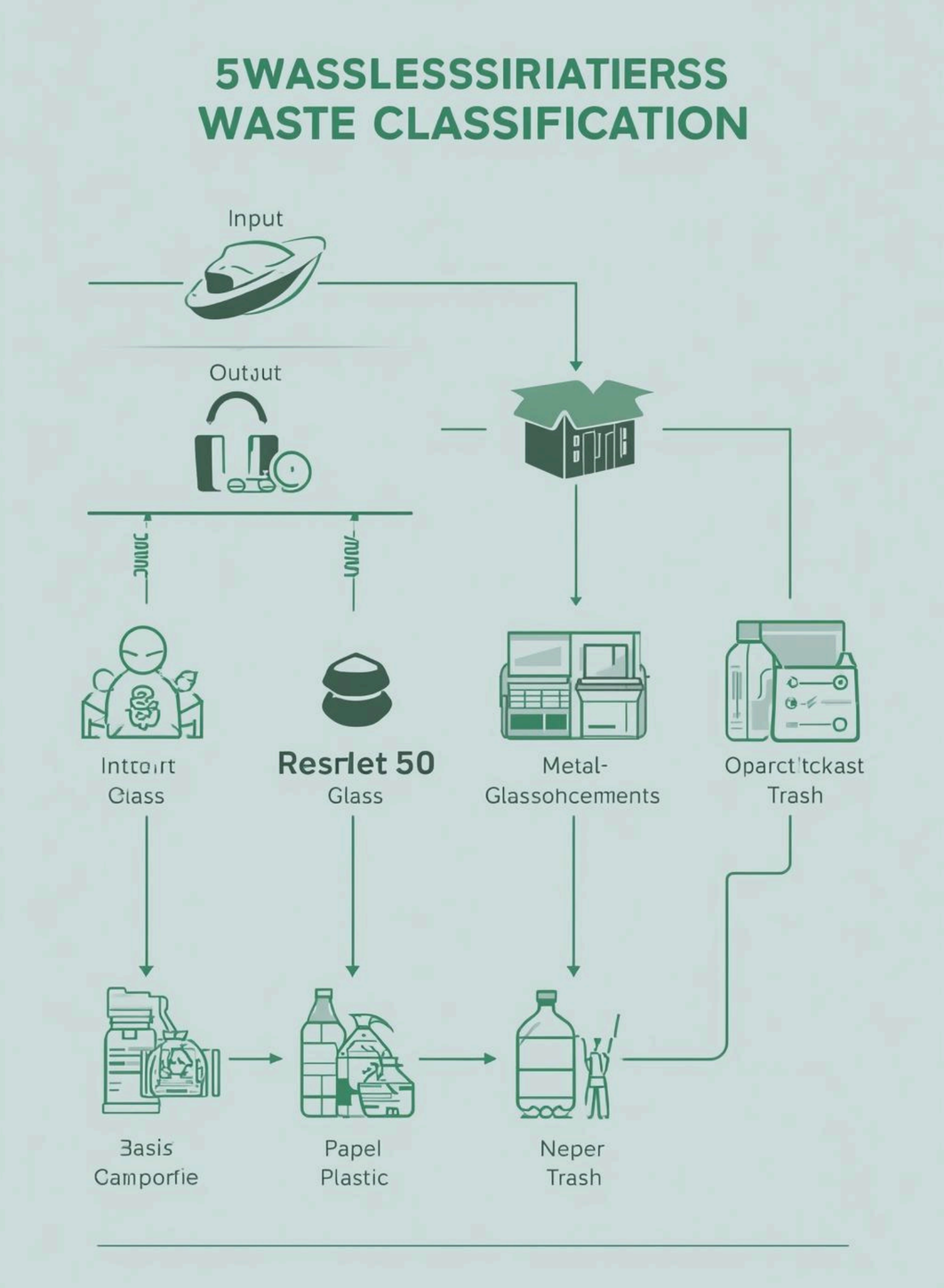
Introduction to Waste Classification

Utilizing ResNet-50 on TrashNet Dataset

This project implements **ResNet-50** for waste classification, trained on the **TrashNet dataset**, integrating adversarial training to enhance model resilience against attacks, ensuring reliable performance in real-world applications.



Importance of Waste Segregation

Environmental benefits of proper sorting

Improper waste segregation leads to **environmental degradation**, increased pollution, and loss of resources. Implementing AI for automatic sorting can significantly enhance efficiency and reduce the negative impact on our planet.

Literature Review

TrashNet Dataset

TrashNet is a widely used dataset for waste classification, containing images of various waste types to train machine learning models.

FGSM & PGD

Fast Gradient Sign Method and Projected Gradient Descent are two adversarial attack techniques demonstrating vulnerabilities in deep learning models under slight perturbations.

Gaps in Research

Few studies combine waste classification with adversarial robustness, highlighting the need for comprehensive research addressing both aspects effectively.

CNNs in Use

Convolutional Neural Networks like ResNet and VGG have shown strong performance in image classification tasks, particularly in waste sorting applications.

Our Project

This project integrates waste classification, adversarial vulnerability testing, and robust training methods to enhance the effectiveness of AI in waste management.

Methodology Overview

Pretrained ResNet-50

We utilized a **pretrained ResNet-50** model to leverage its existing knowledge and enhance our classification accuracy for waste images.

Stage 1: Training

In the first stage, we trained the model on the **TrashNet dataset**, ensuring it learns to differentiate between six waste categories.

Stage 2: Adversarial Testing

The second stage involved applying **FGSM and PGD attacks** to evaluate the model's vulnerability to adversarial perturbations during inference.

Evaluation

Finally, we assessed the model's **performance and robustness** post-adversarial training, ensuring it maintains accuracy against potential adversarial threats.

Dataset Overview

Source

The TrashNet dataset is sourced from GitHub, created by Gary Thung for waste classification purposes.

Train/Val/Test

The dataset is split into training, validation, and test sets with a ratio of 70%, 15%, and 15% respectively.

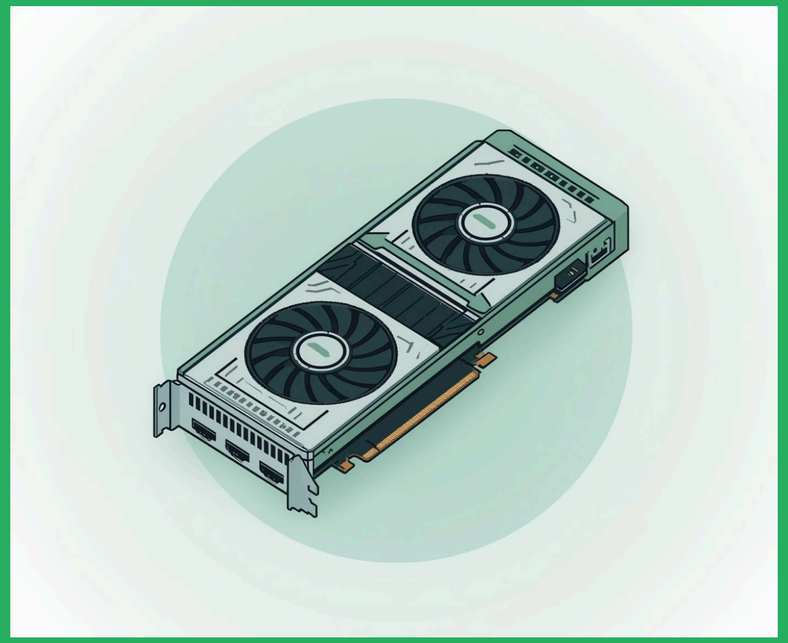
2527 Images

It contains a total of 2527 images categorized into six distinct waste classes, ensuring comprehensive coverage of common trash types.

Augmentations

Various augmentations such as cropping, flipping, rotation, and color jitter are applied to enhance the dataset's variability and robustness.

Implementation Details



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Model

ResNet-50 pretrained on ImageNet, fine-tuned for waste classification.

Optimizer

SGD optimizer utilized with momentum for effective training performance.

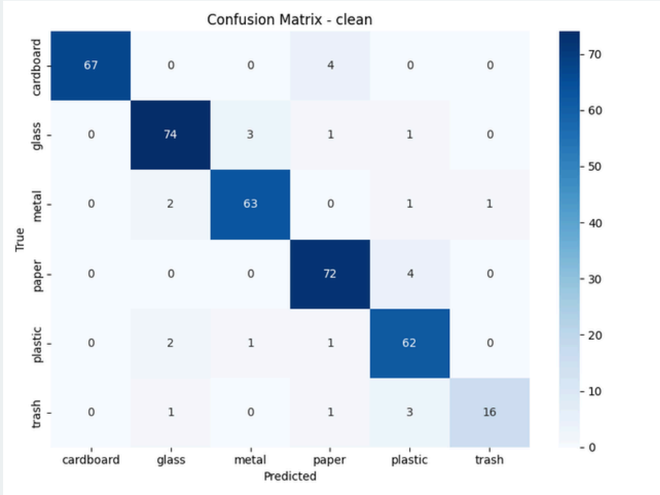
Attack Parameters

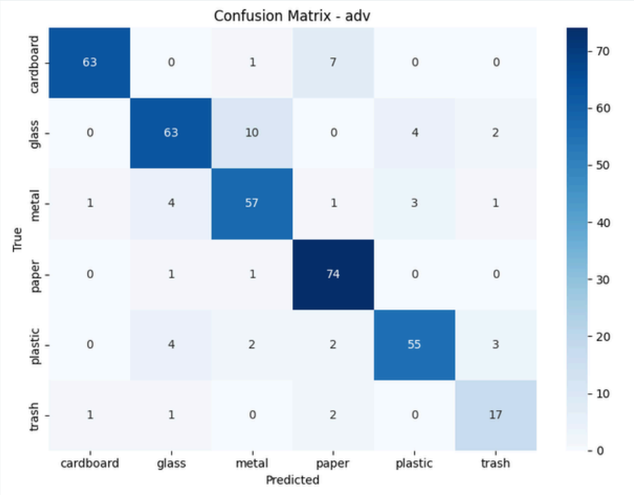
FGSM and PGD settings defined to assess model robustness against adversarial attacks.

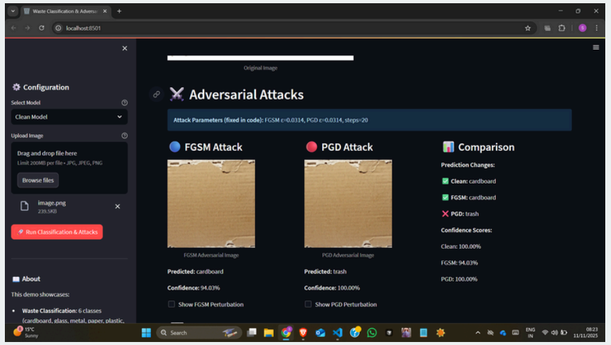
Model Performance Overview

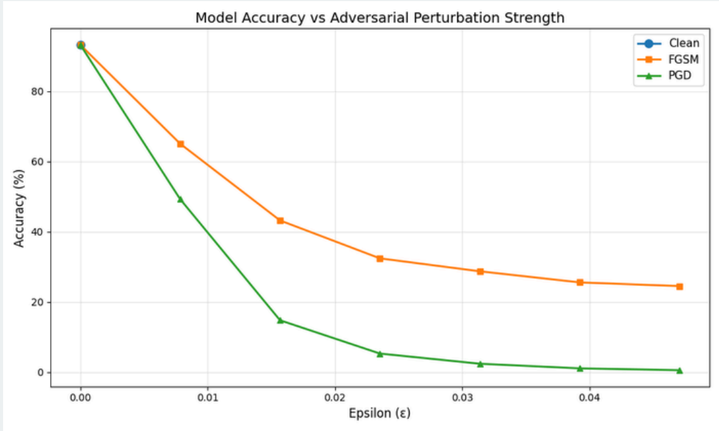
Evaluation of accuracy metrics

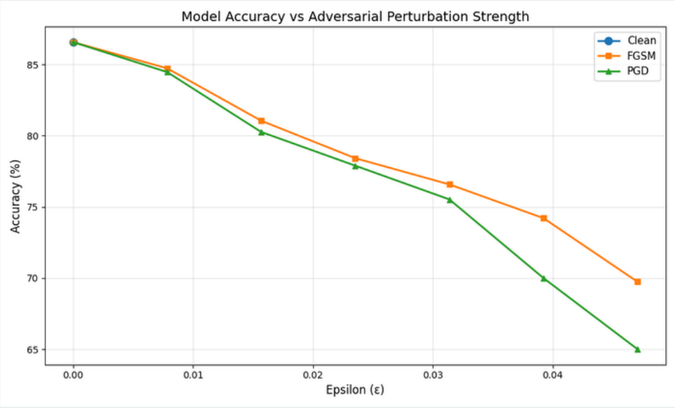
Clean Model Accuracy : 93.16%
FGSM Acc (Clean) : 28.68% , PGD Acc (Clean) : 2.37%
Adversial Model Accuracy : 86.58%
FGSM Acc (Adv) : 76.58% , PGD Acc (Adv) : 75.53%











Conclusion & Future Work

Key findings and next steps in research

Strong Accuracy

ResNet-50 demonstrated strong clean accuracy on the TrashNet dataset, showcasing its ability to classify waste effectively and reliably under normal conditions.

Future Directions

Future work will focus on expanding the dataset, integrating advanced techniques like TRADES, and exploring mobile deployment options to enhance accessibility and usability of the model.